

5 hours examples

Problem 1 (SEGMENTS)

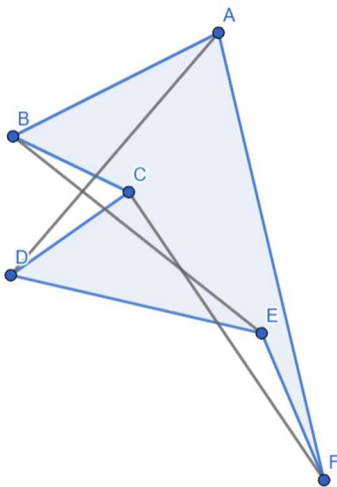
We prove the stronger claim that this is true for any intervals, including $(a_1, b_1]$, $[a_1, b_1)$, or $[a_1, b_1]$. We induct on n .

Base case: $n = 1$. Trivially true.

Inductive step: WLOG, let $a_1 \leq a_2 \leq \dots \leq a_n$. We remove $I_2 \cap I_1$ from I_2 . We now color $I_2, \dots, I_n$ according to the problem conditions. To restore I_1, I_2 , we keep I_2 's color and set I_1 's color to something different.

Problem 2 (TWCH0923)

I don't think this is really an anti-problem. It's more of a free since you just move the points around until the diagonals leave the polygon.



4 clubs; 6 hours

Problem 5 (23PHI1)

We claim $(1, 2)$ and $(2, 1)$ are the only solutions. Both clearly work.

We see that $a^2 - 15ab + b^2 + 25 = 0$. We add $-\frac{1}{17}$ of the expression from $a^2 + ab + b^2$ without changing its value.

$$\begin{aligned} & \frac{16}{17}a^2 + \frac{32}{17}ab + \frac{16}{17}b^2 - \frac{25}{17} \\ &= \frac{1}{17}(4a + 4b - 5)(4a + 4b + 5) \end{aligned}$$

Therefore, for some prime p ,

$$(4a + 4b - 5)(4a + 4b + 5) = 17p$$

Clearly, the only way this is possible is $a + b = 3$.

Problem 6 (10SLA3)

We claim the minimum is $\frac{25}{2}$, achieved by taking $x_{\text{odd}} = \frac{1}{2}; x_{\text{even}} = 0$.

We show that $x_i + x_{i+1} + x_{i+2} \leq 1$ and $x_{i+1} + x_{i+2} + x_{i+3} \leq 1$ implies $x_i x_{i+2} + x_{i+1} x_{i+3} \leq \frac{1}{4}$. This has less conditions than the problem since we don't care about, for example, the value of $x_{i-1} + x_i + x_{i+1}$.

If either one of $x_i + x_{i+1} + x_{i+2}$ or $x_{i+1} + x_{i+2} + x_{i+3}$ are strictly less than 1, we may increase or not change the value of $x_i x_{i+2} + x_{i+1} x_{i+3}$ by increasing x_i or x_{i+3} . Therefore, assume $x_i + x_{i+1} + x_{i+2} = x_{i+1} + x_{i+2} + x_{i+3} = 1 \Rightarrow x_i = x_{i+3}$. Let $x_i = x_{i+3} = t; x_{i+1} = x_{i+2} = 1 - t$. Plugging in and taking AM-GM suffices.

Summing $x_i x_{i+2} + x_{i+1} x_{i+3} \leq \frac{1}{4}$ over all odd i , we are done.

8 clubs; 7 hours

Problem 9 (Z59B00DD)

We do this with strong induction on n .

Base case: $n = 1$. Trivially

Inductive step: By Bertrand's, there is a prime on $2n + 1 \leq p \leq 4n - 1$. For this choice of odd p , we pair $\{2n, p - 2n\}, \{2n - 1, p - 2n + 1\}, \dots$. We are left with $1, \dots, p - 2n + 1$ which can be paired by inductive hypothesis.

Problem 11 (14JMO1)

Let $g(x) = 10x^2 - 5x + 1$; $f(x) = x^2 - 5x + 10$.

$$\min\left(\frac{g(a)}{f(b)}, \frac{g(b)}{f(c)}, \frac{g(c)}{f(a)}\right) \leq abc$$

By Power-Mean,

$$\Leftrightarrow \frac{g(a)g(b)g(c)}{f(b)f(c)f(a)} \leq a^3b^3c^3$$

$$\Leftrightarrow \frac{g(a)}{f(a)} \leq a^3$$

$$\Leftrightarrow 10a^2 - 5a + 1 \leq a^5 - 5a^4 + 10a^3$$

This expands to the binomial theorem, and we are done for $a \geq 1$.

12 clubs; 8.5 hours

Problem 12 (18PTNMB6)

If we let $S = \{1, 2, \dots, 6, 10\}$, note that $\sum_{s \in S} 2^{-s} = \frac{1009}{1024} = \frac{2018}{2048}$. Let $f(n, k)$ be the number of n -tuples with entries from S and summing to k . We use strong induction to show that, for all n, k , we have $f(n, k) \leq 2^k \cdot \left(\frac{2018}{2048}\right)^n$.

Base case I: $n = 0$. $f(n, k) \leq 1 \leq 2^k$ as desired.

Base case II: $k \leq 0$; $n > 0$. $f(n, k) = 0 \leq 2^k \cdot \left(\frac{2018}{2048}\right)^n$ as desired.

Inductive step:

$$f(n, k) = \sum_{s \in S} f(n-1, k-s) \leq \sum_{s \in S} 2^{k-s} \cdot \left(\frac{2018}{2048}\right)^{n-1} = \sum_{s \in S} 2^k \cdot \left(\frac{2018}{2048}\right)^n$$

Problem 15 (ZB2C2EFD)

We manipulate $9(a-b)(a^2-b^2)(a^3-b^3) = (a^2-2b^2+ab)^3 + (2a^2-b^2-ab)^3$.

Problem is now trivial by Fermat's Last Theorem.

16 clubs; 10 hours

Problem 16 (Z9503B7B)

Answer no. FTSOC possible.

Sum s already even. For s to be semiprime, $s = 2p$ for other prime p . Average of our two primes must be another prime, contradicting consecutive.

Problem 18 (Z331EC32)

We scale down the construction for R by $\frac{1}{2}$ then copy-paste it four times to reconstruct R .

20 clubs; 12 hours

Problem 20 (10BRMO1)

We claim no.

Have the graph of the friendships be a complete binary tree with height 1435.

Of the 2^{1434} friends at layer 1435, each of them must be within $1434 \cdot 2010$ of the root in the line. Contradiction by Pigeonhole.

Problem 23 (CF1485D)

Note that $\text{lcm}(1, \dots, 16) = 720720 < 10^6$. We fill each cell (i, j) such that $2 \mid i + j$ as 720720 and every other cell with value k as $720720 - k^4$ to suffice.

<https://codeforces.com/contest/1485/submission/309905208>

24 clubs; 14 hours

Problem 24 (09ELMO4)

We first draw a path “snaking” through $\left(\frac{1}{2n}, 0\right) \rightarrow \left(\frac{1}{2n}, 1\right) \rightarrow \left(\frac{3}{2n}, 1\right) \rightarrow \left(\frac{3}{2n}, 0\right) \rightarrow \left(\frac{5}{2n}, 0\right) \rightarrow \left(\frac{5}{2n}, 1\right) \rightarrow \dots$. The total length of the path is $n + 1 - \frac{1}{n}$. For a given point, we drop the altitude from it to the closest point on the graph. We add this altitude to the segment twice for length at most $\frac{1}{2n} \cdot 2 = \frac{1}{n}$. The total length is now at most $n + 1 - \frac{1}{n} + \frac{1}{n} \cdot n^2 = 2n + 1 - \frac{1}{n}$.

Problem 25 (Z52110CF)

There's a heptomino with a hole in it lmao.

28 clubs; 16 hours